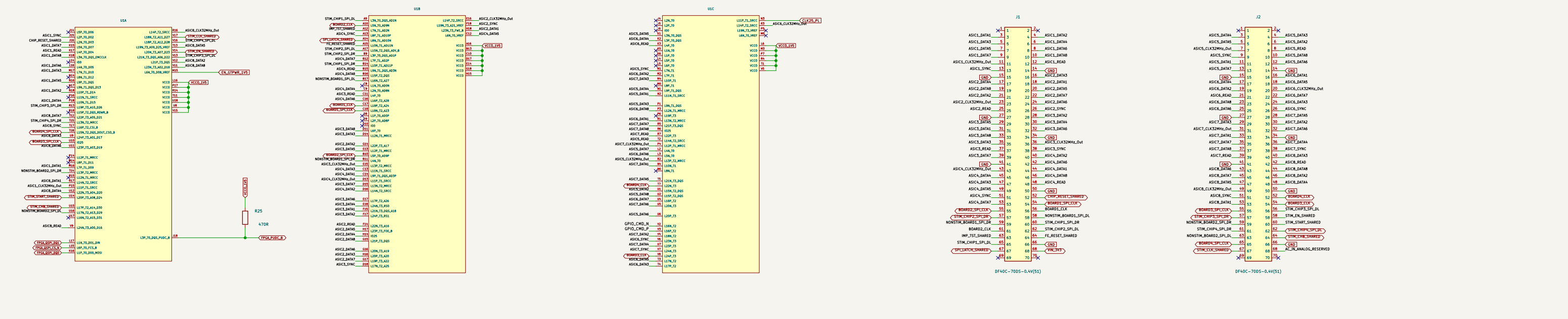
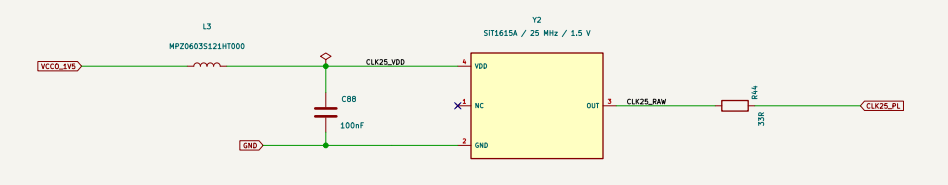
01. FPGA I/O + ASIC INTERFACES + 25 MHz CLOCK | Trenz p5



1. Function: U1 / J1 / J2: Route 116 active signals for eight 1.5 V ASICs through two 70-contact mezzanines with 7 grounds each. J1.68 carries U16-regulated VIN_3V3; J2.68 stays reserved analog; corner contacts 1, 2, 69, 70 are unconnected.

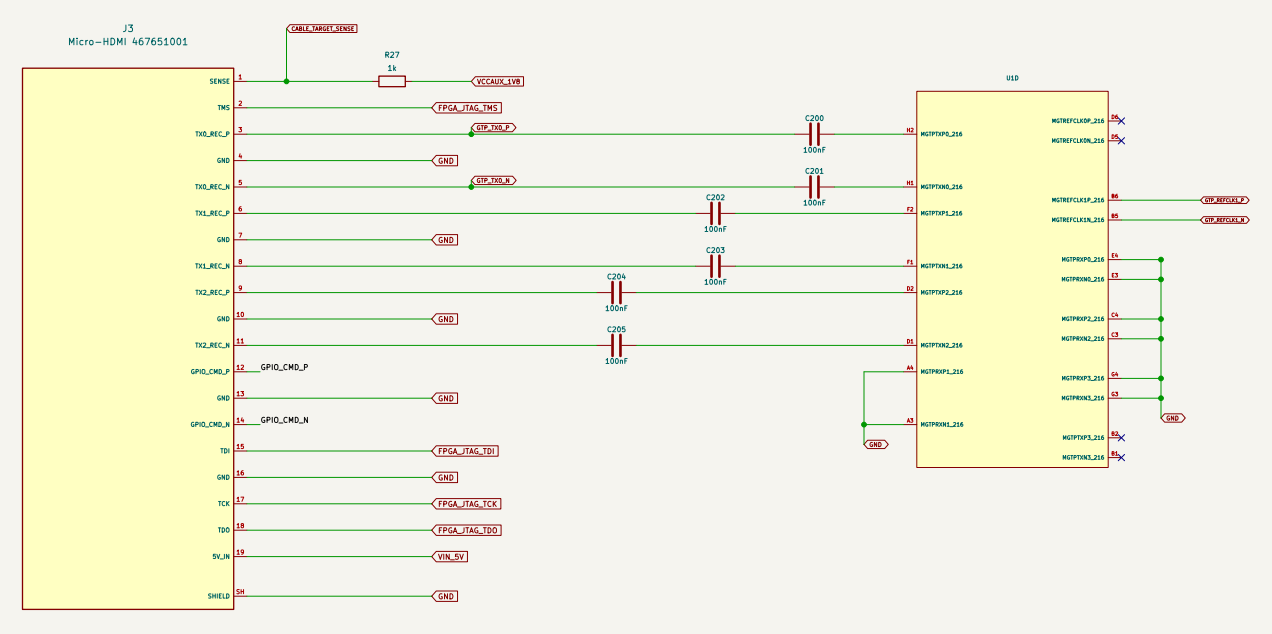
2. Source / changes: Codex custom ASIC interface. V8 FPGA-side pin remap (Howard approved 2 Nov 2026): ASIC groups moved to the bank nearest their connector, each returned clock on a clock-capable pin; Vivado check pending. V11 change (Howard approved 2 Nov 2026): J1/J2 DF40C-60DS to DF40C-70DS and contact map changed and reordered within each ASIC group except ASIC1. FPGA pins unchanged, because the header corner contacts are not energized: Hirose DF40 catalog Jun 2024 p.8 and p.14 Note 3, guideline ETAD-H1015-00 sec 1.2.1. https://www.hirose.com/en/product/series/DF40.
Mating DF40C-70DP on the unbuilt LDO board and the contact map on that board are unverified. ASIC_REFERENCE.md sections2-5; Chip_FPGA interface.pptx slides13-17/25-28; Pins.xlsx. Trenz p5 I/O bank use adapted to ASIC allocation. https://www.trenz-electronic.de/trenzdownloads/Trenz_Electronic/Modules_and_Module_Carriers/3x4/TE0714/REV04/Documents/SCH-TE0714-04-42I-7-L.PDF ASIC silicon timing/rail limits remain unqualified.



1. Function: Y2:25MHz fabric clock drives N3 SRCC at1.5V. L3 filters1.5V; C88=100nF; R44=33R output damping.

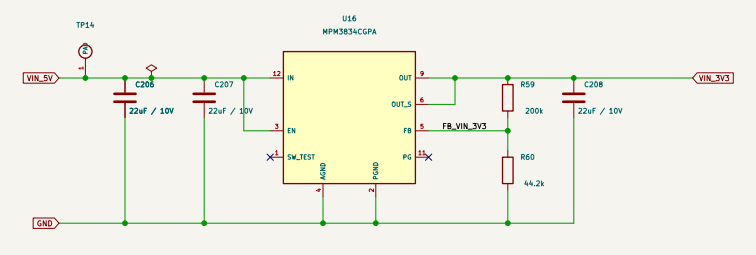
2. Source / changes: Trenz p5 U8/L2/C24/R23 structure adapted: SIT1615 1.5V replaces SIT8008 1.8V; CLK25 uses N3 SRCC. C88 retained100n per SiTime; R44 exact Trenz RC0201FR-0733RL restored. https://www.trenz-electronic.de/trenzdownloads/Trenz_Electronic/Modules_and_Module_Carriers/3x4/TE0714/REV04/Documents/SCH-TE0714-04-42I-7-L.PDF https://www.sitime.com/datasheet/SIT1615 Timing/firmware constraints remain unverified.

02. GTP CHANNELS + 125 MHz REFERENCE | Trenz p6



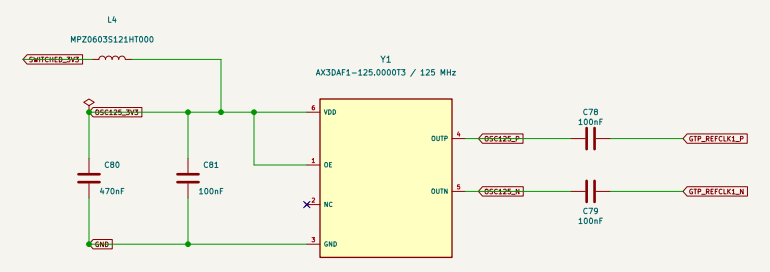
1. Function: J3 carries three GT recording TX pairs, regular-I/O commands, JTAG and 5 V power (contact 19 -> VIN_5V -> U16). TX0 connects to XEM; TX1/TX2 end open at carrier.

2. Source / changes: Codex custom HDMI map adapted from V3; 9/11 add TX2 via C204/C205; 12/14 move from GTP RX1 to U1 V3/V2 regular I/O.
AMD DS181 v1.27.1 Table 50: CEXT=100 nF; C200-C205 are 0201 Murata GRM033R61A104KE15D (V9: 0402 to 0201 for pair symmetry; UG482 allows 0402 or 0201). https://docs.amd.com/api/khub/documents/iAkxxTOk96ANLJqYf2hgrQ/content
Unused RX1 A4/A3 grounded: AMD UG482 v1.9 Table 5-6 pp.223-224. https://docs.amd.com/api/khub/documents/Xk_zMbXsJj92suJ4_0Akkg/content
1.5 V Bank34 command proposal: DIFF_SSTL15, IN_TERM=UNTUNED_SPLIT_50: AMD UG471 v1.10 Table 1-7. https://docs.amd.com/api/khub/documents/lbGcnPFe6ef19RHma_Y~lA/content
Carrier needs compatible regular I/O and VIO1=1.5 V. TX0 connects to XEM; TX1/TX2 are future carrier NC. Pin/clock/protocol, coupling, cable SI, shell bonding and unpowered-target protection remain unverified. V5 changes contact 19 from 3.3 V to 5 V (carrier TPS2553 port feed).



1. Function: U16 converts the 5 V cable feed on J3 contact 19 (VIN_5V, about 4.4-5.0 V after cable drop) into VIN_3V3, which supplies every existing regulator, logic part and J1.68. C206/C207 input, C208 local output; EN tied to VIN_5V; PG unused; TP14 probes the cable voltage at the frontend.

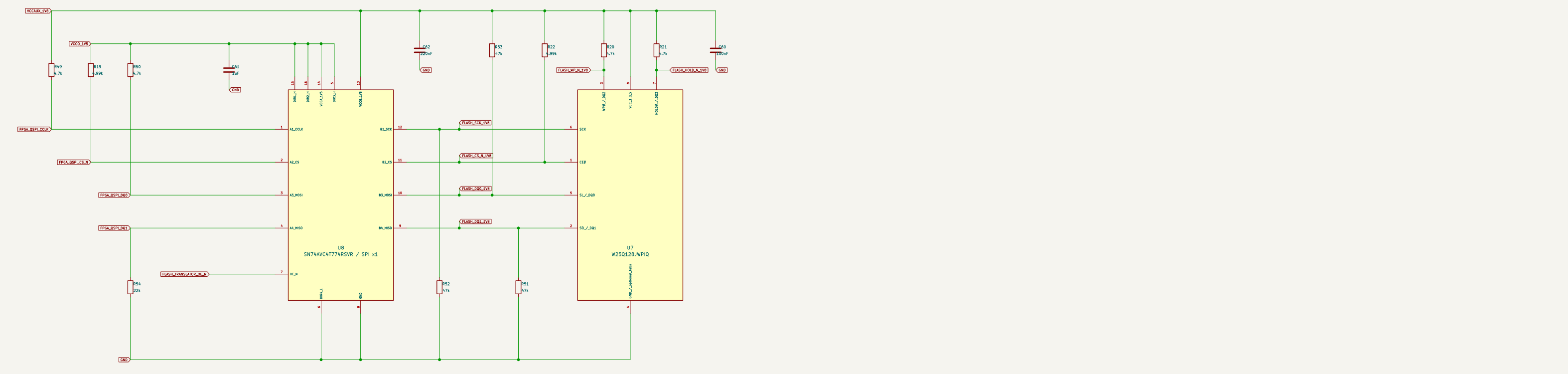
2. Source / changes: MPS MPM3834C Rev 1.01 (6/14/2022) Fig. 4 p17 typical application; Table 1 p14 (3.3 V: R1=200k, R2=44.2k); input/output capacitor guidance p14;
VIN 2.75-6 V operating, 6.5 V abs max p3. https://www.monolithicpower.com/en/documentview/productdocument/index/version/2/document_type/Datasheet/lang/en/sku/MPM3834CGPA/
Adapted by Claude Code: R59/R60 = Table 1 values (VOUT = 0.6 V x (1 + 200k/44.2k) = 3.31 V); CFF not stuffed as Fig. 4; Fig. 4 R3/R4 EN divider replaced by direct EN tie (as U2);
input 2 x 22 uF 10 V (Fig. 4: one 22 uF); TP14 added. Unverified: cable hot-plug overshoot vs 6.5 V abs max (cable resistance should damp it; measure), real frontend current, layout and thermal.



1. Function: Y1 supplies125MHz LVDS through C78/C79=100nF to REFCLK1P/N. L4 filters switched3.3V; C80=470nF and C81=100nF bypass it. The REFCLK input's internal 100R is the only load; OE HIGH, pin2 NC.

2. Source / changes: Trenz TE0714-04-42I-7-L REV04 p6;125MHz LVDS U2, L1 ferrite, C10=470nF, R1=100R and C134/C135=100nF coupling. https://www.trenz-electronic.de/trenzdownloads/Trenz_Electronic/Modules_and_Module_Carriers/3x4/TE0714/REV04/Documents/SCH-TE0714-04-42I-7-L.PDF Adapted by Codex: Y1=AX3DAF1-125.0000T3 (exact Trenz oscillator MPN undisclosed). C80=470nF, C81 10uF->100nF; added L4. External R56 removed per UG482 Fig. 5-8 p226 (100nF caps only); with it, AX3 swing is 250-450mV against DS181 350mV minimum. Abracon AX3 rev25Feb2022 pp3,9,12-13 requires active-HIGH OE and shows100nF local bypass. https://abracon.com/datasheets/AX3.pdf REFCLK1 B6/B5 matches Trenz; P-to-P/N-to-N retained. AMD UG482 v1.9 pp225-227,235; LVDS AC-coupled reference. https://docs.amd.com/v/u/en-US/ug482_7Series_GTP_Transceivers Clock jitter, filtered-rail noise, layout and hardware startup remain unqualified. R56 uses the same100R0.1%25ppm candidate as R16; exact source MPN is undisclosed.

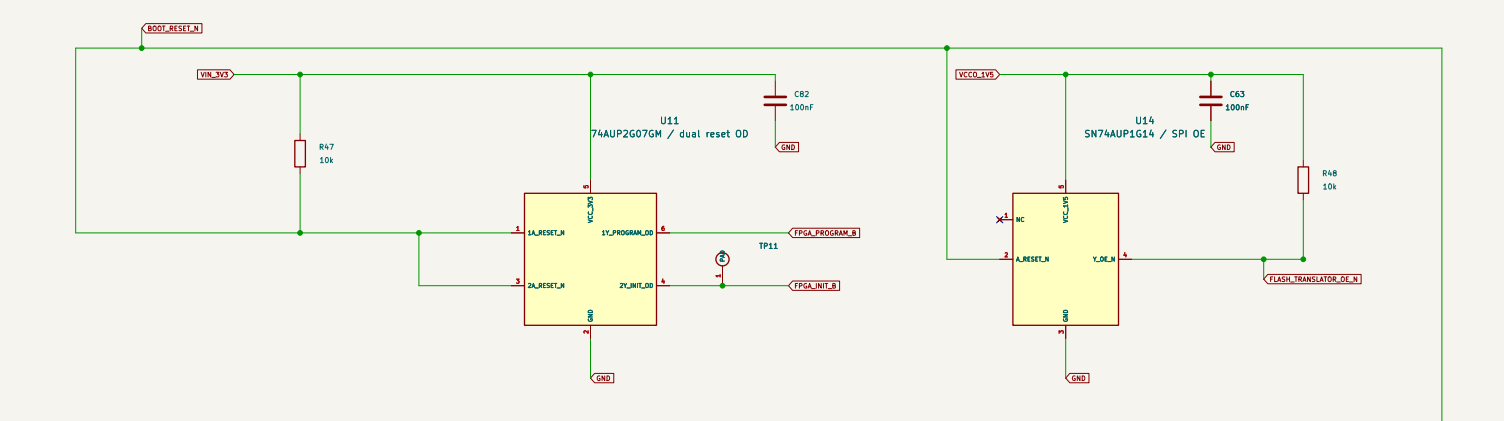
03. FLASH + SPI TRANSLATION | Trenz p6 adaptation



1. Function: U7/U8: 16MiB 1.8V SPI NOR flash (Winbond W25Q128JWPIQ, 6x5mm WSON-8) and SN74AVC4T774RSV SPIx1 translation. Bank14 remains1.5V; initialCONFIGRATE3MHz/QE=1 fixed/3-byte addresses.

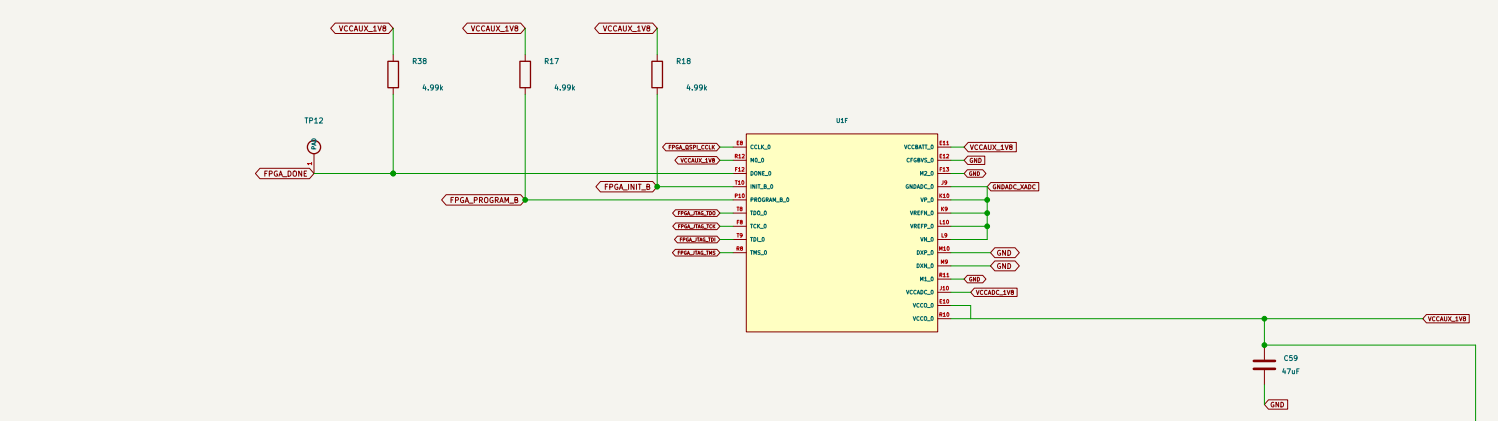
2. Source / changes: V8 replaces unsubstantiated V6/V7 W25Q128JWUUIQ with Winbond W25Q128JWPIQ; W25Q128JW Rev.H 17Apr2023, pin table p5, WSON 6x5 outline p68, ordering pp75-76. https://mm.digikey.com/Volume0/opasdata/d220001/medias/docus/8919/256_W25Q128JW.pdf TI translator: https://www.ti.com/lit/ds/symlink/sn74avc4t774.pdf R19/R22 4.99k; pullups 1.8V. Codex/Claude adaptation of Trenz boot structure. Q option fixes QE=1, so WP#/HOLD# functions are off. Land pattern, boot timing and Vivado programming support unverified.

04. CONFIGURATION + RESET OUTPUTS | Trenz p7 notes



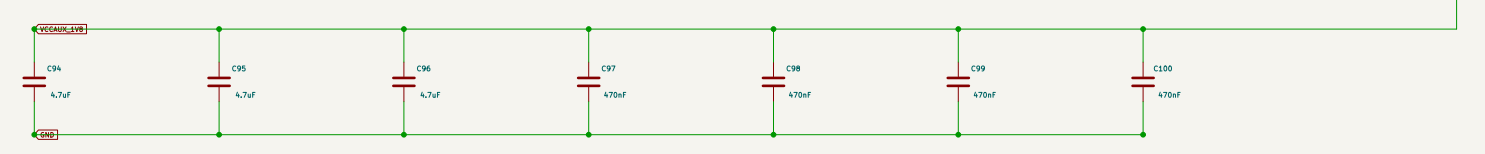
1. Function: U11 holds PROGRAM_B and INIT_B LOW through two independent open-drain outputs; R17/R18 release them to 1.8 V. U14 drives the 1.5 V active-low U8 enable. C82/C63 are local bypass.

2. Source / changes: Correspondence: Trenz p7 D1/D2 reset-hold functions -> U11; U14 is added for our U8 flash translator, not Trenz boot-mode inverter U9. V64 electrical network unchanged. Codex voltage adaptation of Trenz p7 D1/D2 reset function; inherited V61 agreed flash/U8 support retained. Trenz diode/common3.3V solution is not used across our1.8V/1.5V domains. https://www.trenz-electronic.de/trenzdownloads/Trenz_Electronic/Modules_and_Module_Carriers/3x4/TE0714/REV04/Documents/SCH-TE0714-04-42I-7-L.PDF
https://assets.nexperia.com/documents/data-sheet/74AUP2G07.pdf https://www.ti.com/lit/ds/symlink/sn74aup1g14.pdf



1. Function: U1 configuration: Master SPI001,CFGBVS GND, Bank0/JTAG1.8V; ASIC banks1.5V. PROGRAM_B and INIT_B reset independently. VCCBATT E11 now AUX1.8V.

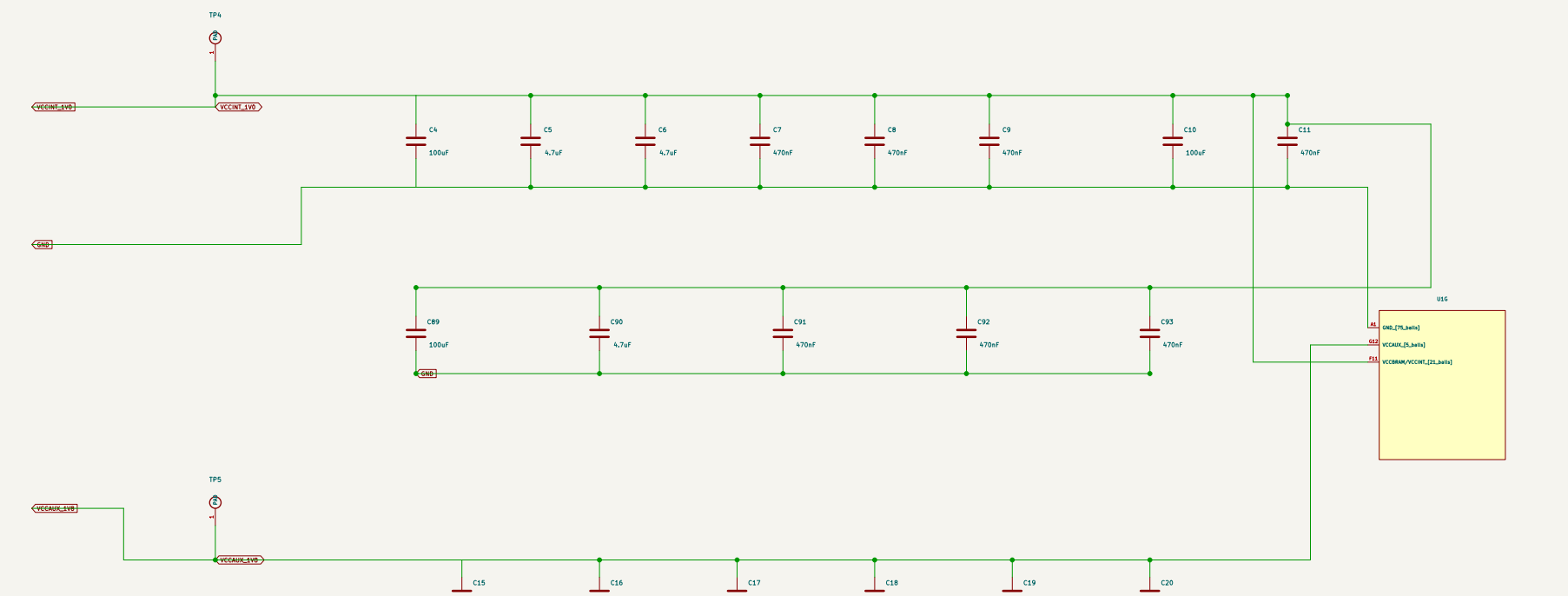
2. Source / changes: Trenz p7 values restored where compatible: R17/R18/R19/R22=4.99k; E11 GND->AUX. Mode ties/1.8V rails and x1 flash remain agreed adaptations.
https://www.trenz-electronic.de/trenzdownloads/Trenz_Electronic/Modules_and_Module_Carriers/3x4/TE0714/REV04/Documents/SCH-TE0714-04-42I-7-L.PDF AMD UG470v1.17 Table2-4 permits AUX for unused BBRAM.
https://docs.amd.com/api/khub/documents/FOs3lXmlcWxBhTlFxVKyGA/content



1. Function: Bank0 bypass C59,C94-C100:47uF+3x4.7uF+4x470nF at1.8V, separate from1.5V Bank14.

2. Source / changes: Codex split of Trenz p7 shared Bank0/14 network to preserve agreed voltage domains; addedC94-C100. C59 exact Trenz CC0805MKX5R5BB476.
https://www.trenz-electronic.de/trenzdownloads/Trenz_Electronic/Modules_and_Module_Carriers/3x4/TE0714/REV04/Documents/SCH-TE0714-04-42I-7-L.PDF Duplication adds startup charge; effective capacitance and physical placement require validation.

05. FPGA SUPPLY PINS + BYPASS + AUX | Trenz p7

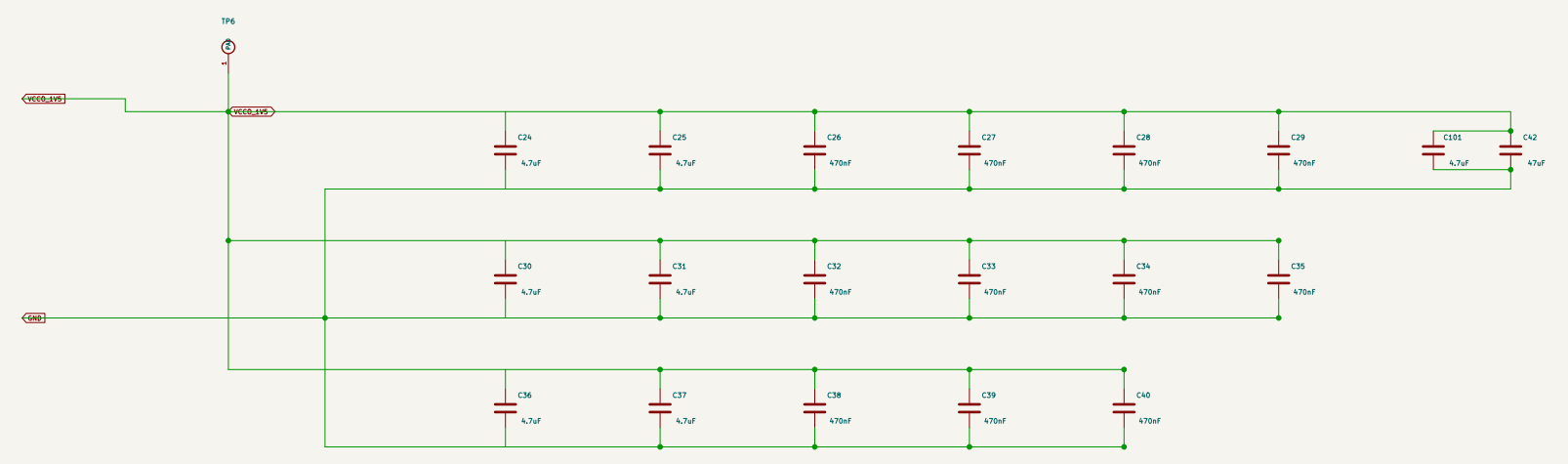


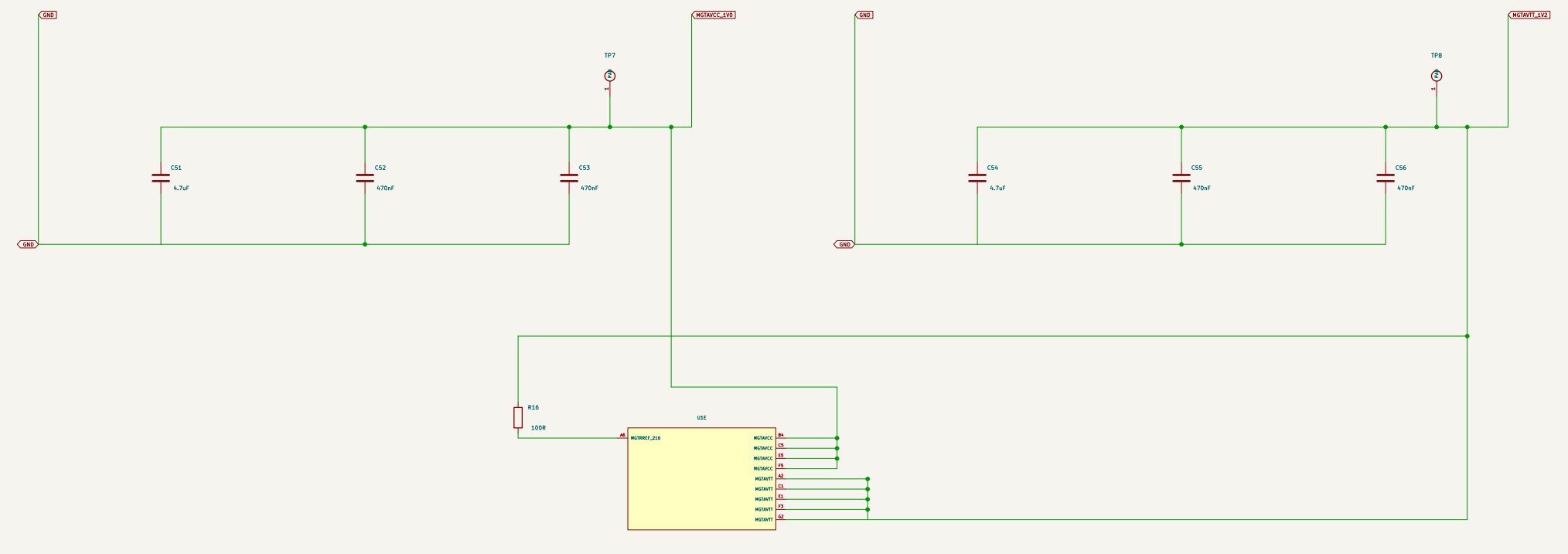
1. Function: U1 core/BRAM bypass: C4-C11,C89-C93 now3x100uF+3x4.7uF+7x470nF on1.0V (317.39uF nominal).

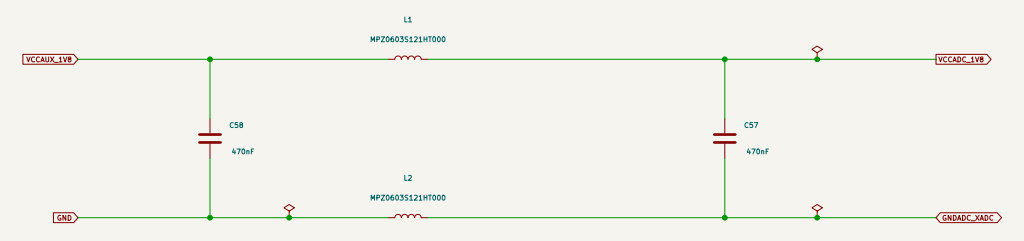
2. Source / changes: Restored Trenz TE0714 REV04 p7 core/BRAM group: 100uF C8/C54/C56;4.7uF C43/C48/C55;470nF C52/C49/C50/C51/C13/C11/C27. Our C10 47u->100u, added C89-C93. Adapted C4/C10/C89 from 1206 (10%) to 0805 Murata GRM21BR60J107ME15L (100uF 6.3V X5R 20%, https://www.mouser.com/datasheet/2/281/1/GRM21BR60J107ME15_01A-1986857.pdf) for V6 size reduction. https://www.trenz-electronic.de/trenzdownloads/Trenz_Electronic/Modules_and_Module_Carriers/3x4/TE0714/REV04/Documents/SCH-TE0714-04-42I-7-L.PDF DC bias, layout, PDN and added startup charge remain unverified.

1. Function: U1 AUX bypass C15-C20:47uF+2x4.7uF+3x470nF on1.8V.

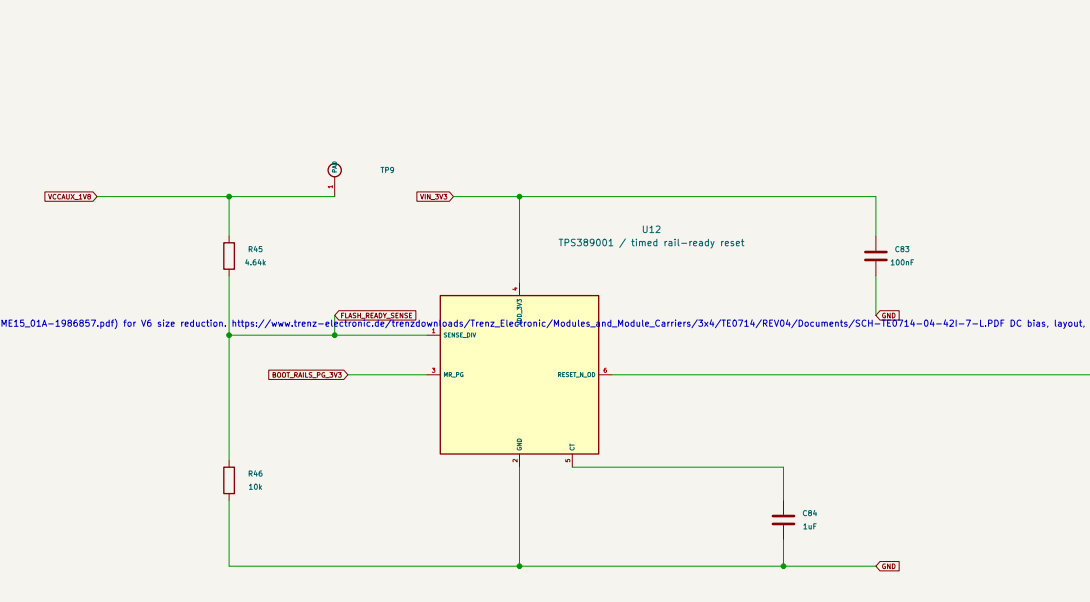
2. Source / changes: Trenz p7 nominal network retained; C15 exact Yageo CC0805MKX5R5BB476, compatible tolerance/package restored. https://www.trenz-electronic.de/trenzdownloads/Trenz_Electronic/Modules_and_Module_Carriers/3x4/TE0714/REV04/Documents/SCH-TE0714-04-42I-7-L.PDF
Capacitor bias/layout remain unverified.





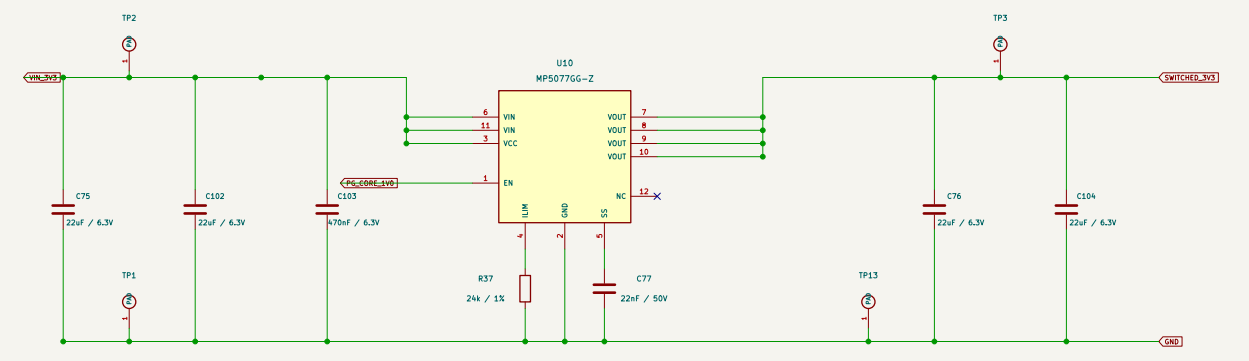


06. REGULATORS + READINESS SUPERVISOR | Trenz p8 notes



1. Function: U12: Sense actual AUX/Bank0/flash1.8V through R45/R46. Final GTP AVCC PG and1.5V PG gate timed reset. C84 sets delay.

2. Source / changes: Correspondence: Trenz p9 U23 monitoring/timing role -> U12 TPS389001. This remains our adaptation, not an unchanged STM6710 circuit. V64 values/connections unchanged. Codex TPS389001 adaptation retains agreed V61 threshold network rather than Trenz p9 STM6710 supervisor. PG input now follows AVTT->AVCC ordering. https://www.trenz-electronic.de/trenzdownloads/Trenz_Electronic/Modules_and_Module_Carriers/3x4/TE0714/REV04/Documents/SCH-TE0714-04-42I-7-L.PDF https://www.ti.com/lit/ds/symlink/tps3890.pdf Threshold tolerances, startup and power-down remain unverified.

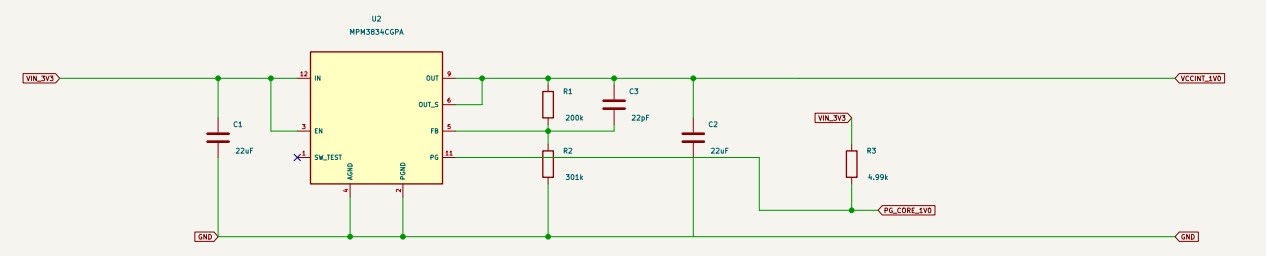


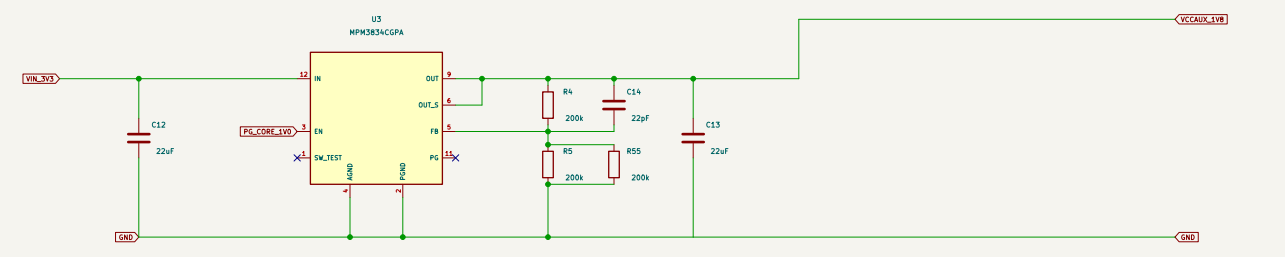
1. Function: MP5077 enables the local3.3V clock/support branch after core PG. VIN_3V3 from U16 (5 V cable input) feeds FPGA buck inputs and GTP LDO BIAS directly. The carrier port switch limits cable current.

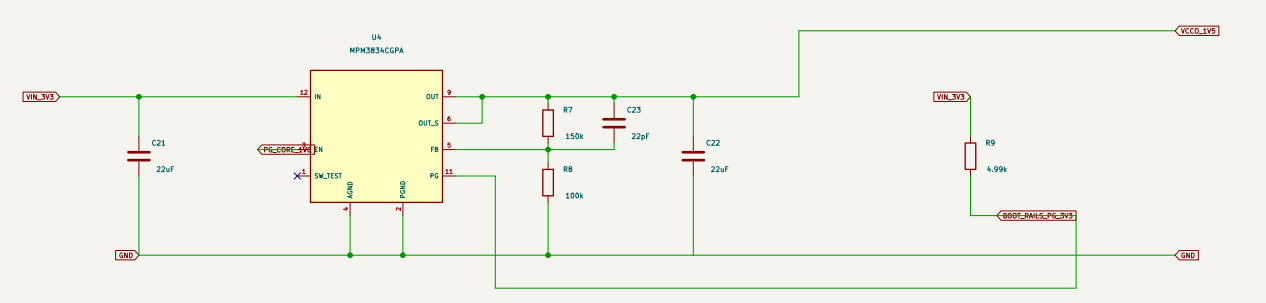
2. Source / changes: Trenz TE0714-04-42I-7-L REV04(20Dec2022),p9 Q1/R52/C1/input-output bypass copied: https://www.trenz-electronic.de/trenzdownloads/Trenz_Electronic/Modules_and_Module_Carriers/3x4/TE0714/REV04/Documents/SCH-TE0714-04-42I-7-L.PDF MPS MP5077 Rev1.2(15Aug2023),p2 pinout,pp11-14 limit/SS/land pattern: https://www.monolithicpower.com/en/documentview/productdocument/index/version/2/document_type/Datasheet/lang/EN/sku/MP5077 V63 replaces whole-board TPS259470A; R34/R35/R36 removed. R37=24k1%,C77=22nF50VX7R; C75/C76=22uF; added C102/C104=22uF andC103=470nF. Nominal branch ILIM3.42A and source SS label3.5ms are not whole-board cable ratings. TP1/TP13 GND,TP2 rawVIN andTP3 branch retained. Exact Trenz passive MPNs not disclosed; source-matched candidates/pending selections identified in fields. Startup,cable loss,source/branch interactions and thermal behavior remain unqualified.

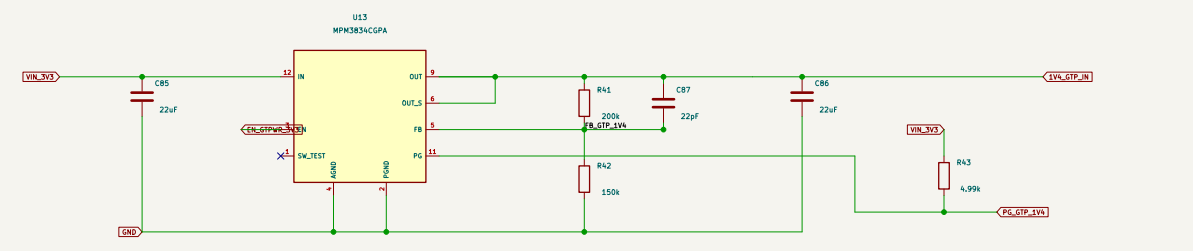
1. Function: TP1-TP13: Expose ground, U16 VIN_3V3, switched3.3V and local rails/status. Virtual power flags declare sources to ERC.

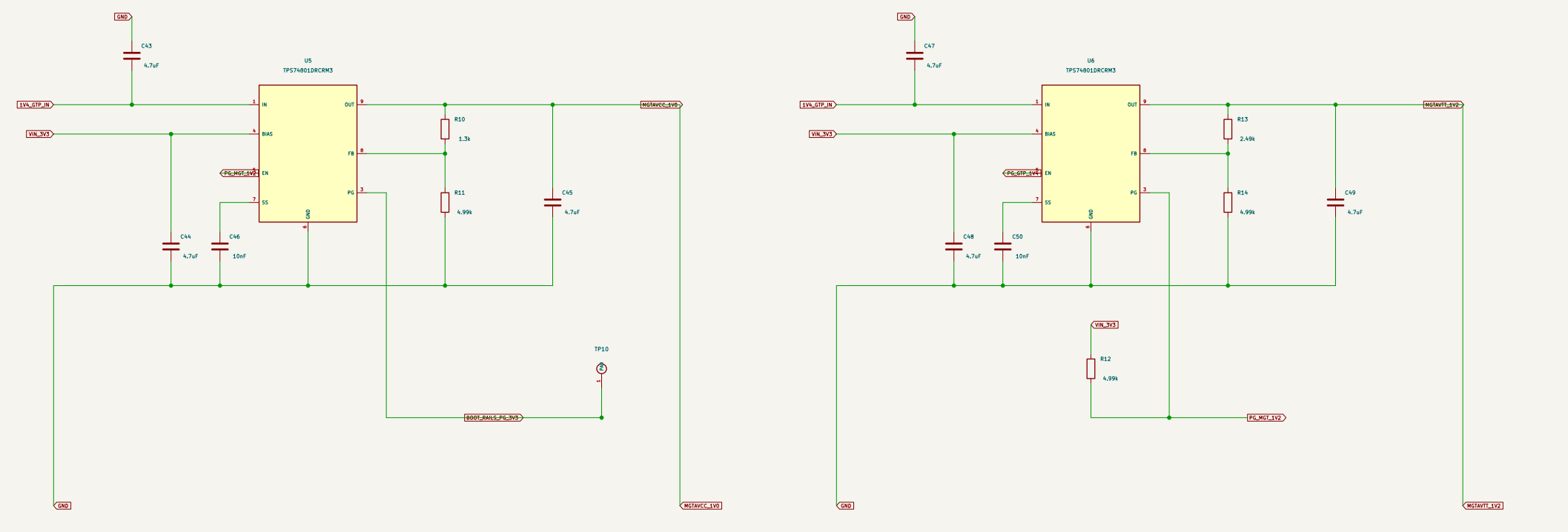
2. Source / changes: Codex debug additions retained;13 custom0.6mm pads versus Trenz21 pads0.8mm. TP10 now finalAVCC-PG; TP3 switchedbranch. No physical testpad datasheet; no V65 PCB. Probe loading/access unverified.

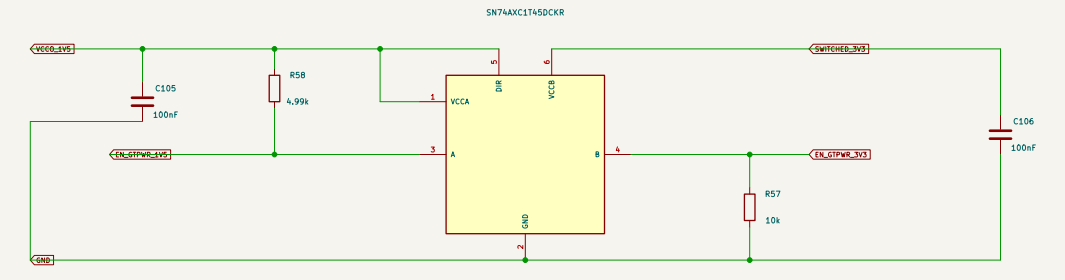












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Size: User | Date: 2026-10-19 | **Rev: V54**
KiCad E.D.A. 10.0.6 | Id: 1/1